

Mikhail Razakov

Software Developer

JUNIOR / INTERNSHIP

compilers · .NET · C++ · verification

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PROFILE

I build bounded software components, define their correctness conditions explicitly, and carry them through to reproducible verification.

KEY EVIDENCE

1,579 tests · 0 failures

Public verification manifest dated August 2, 2026: 1,579 tests, zero failures; nine packages and clean-consumer scenarios.

1st of 49

1st among 49 submitted solutions; the only 5,0/5,0 score and 104/104 official tests in the LLVM compiler track selection.

Correctness verification

Exact-oracle, differential, and metamorphic testing, replay/reduction, and conservative fail-closed decisions.

EXPERIENCE

MCST — Compiler Engineering Intern

July — August 2026

LLVM · C++23 · remote

- Developing an LLVM optimization pass.
- Implemented cycle-detecting RPO, Dijkstra, Dinic maximum flow, and Tarjan SCC over a shared C++23 graph model; CMake build with input/output checks and a Python test runner.

[GitHub](#)

UniversalToolchain/Wist2 — Creator and Primary Developer

2024 — present

.NET SDK · language platforms · runtime

- Designed language composition from independently versioned packages: typed artifacts, capabilities/conflicts, deterministic routes, pass ordering, and exact manifest/lock binding.
- Implemented interpreter/CIL execution, callable-aware SSA, and verifier-gated transformations; unresolved configuration and semantic states fail closed.
- Public verification manifest dated August 2, 2026: 1,579 tests, zero failures; nine packages and clean-consumer scenarios.

[GitHub](#)[Documentation](#)[Verification](#)

SELECTED PROJECTS

PS-form Memory Dependence Analyzer

C17 · program analysis

- C17 analyzer for parametric memory-access dependence with conservative yes/no/maybe outcomes.
- Symbolic normalization, residue/GCD abstractions, exact affine analysis, and cost-guarded fallbacks; a harness with an exact oracle, metamorphic WA hunting, and model-guided TL hunting.

[GitHub](#)

VpnMediator

.NET backend · state · recovery

- Subscription-service backend with an explicit access lifecycle, idempotent operations, and verifiable recovery.

SKILLS

Languages	C#/.NET, C++23, C17, Python, Rust
Compilers	IR/SSA, interpreter/CIL, LLVM, program analysis
Backend	ASP.NET Core, PostgreSQL/SQLite, migrations, recovery
Verification	exact oracles, differential/metamorphic testing, CI, sanitizers

EDUCATION AND RECOGNITION

HSE University — Software Engineering program, since 2026.

HSE Vysshaya Proba prize-winner in competitive and industrial programming; two-time first-degree diploma recipient at the MEPhI Junior competition.